Decision Boundaries

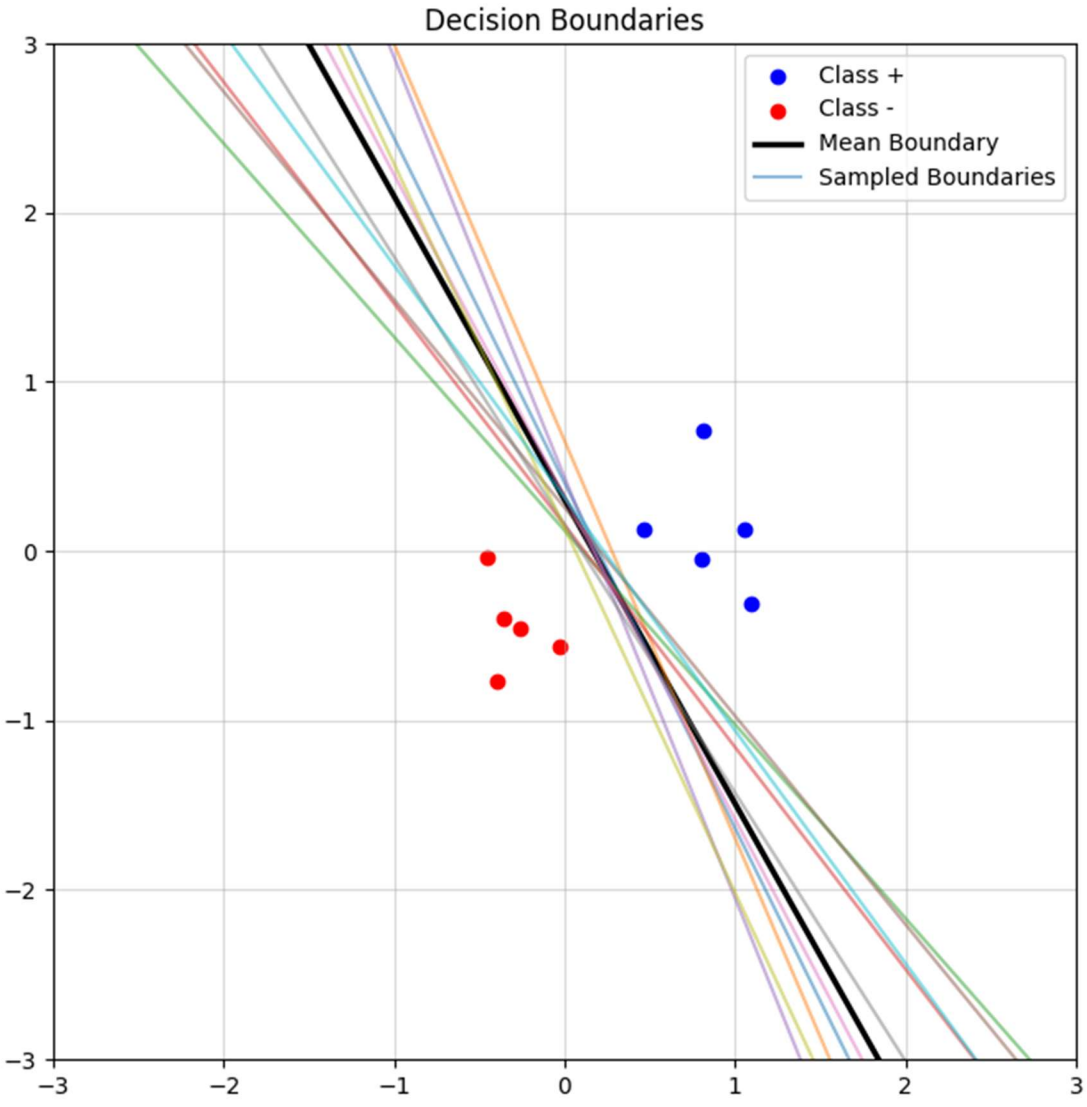
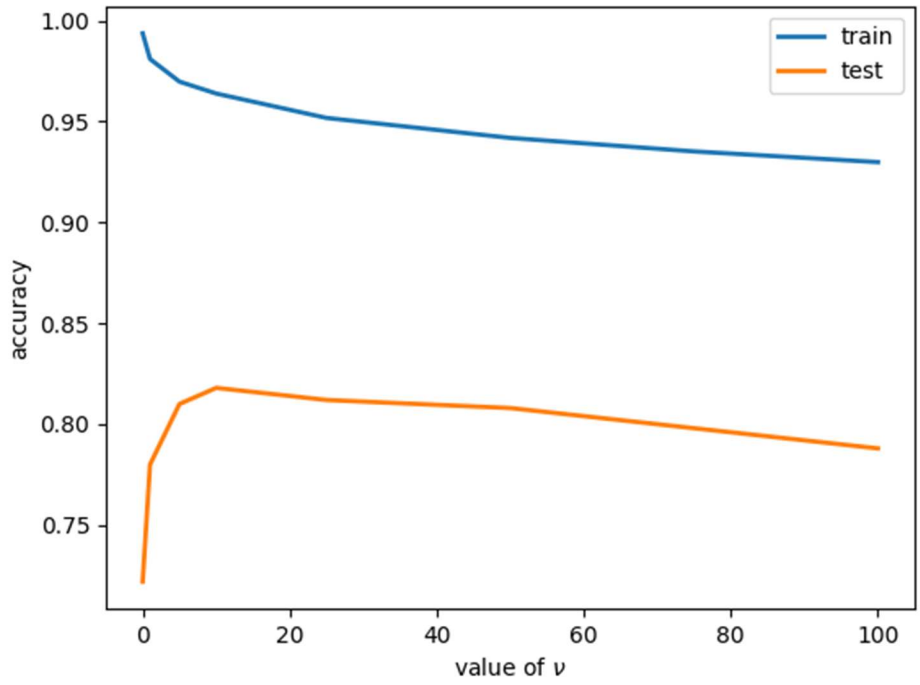
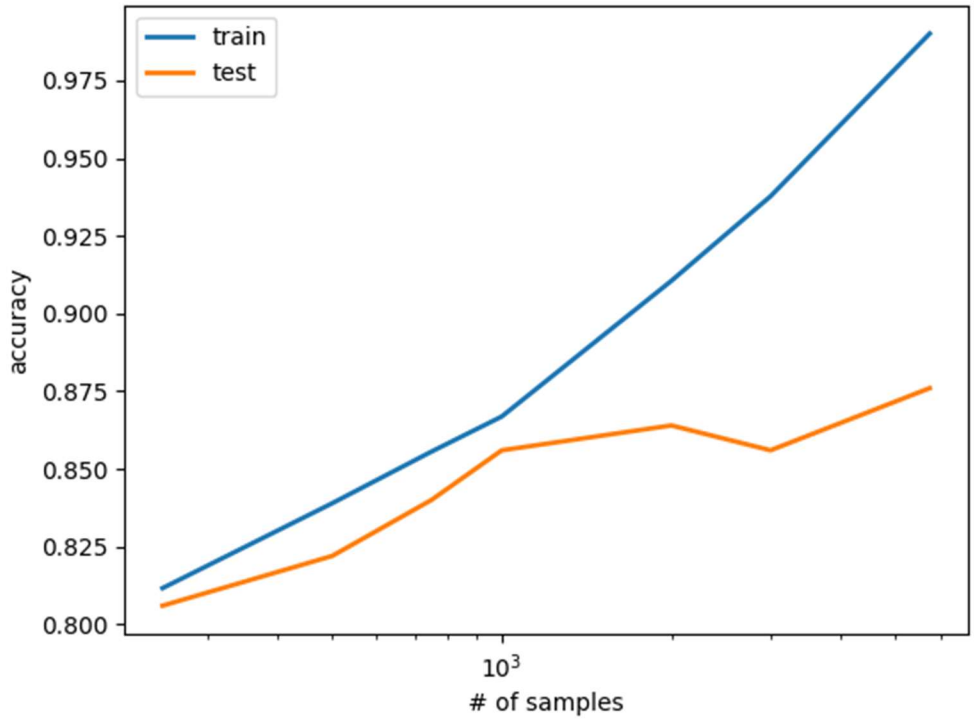
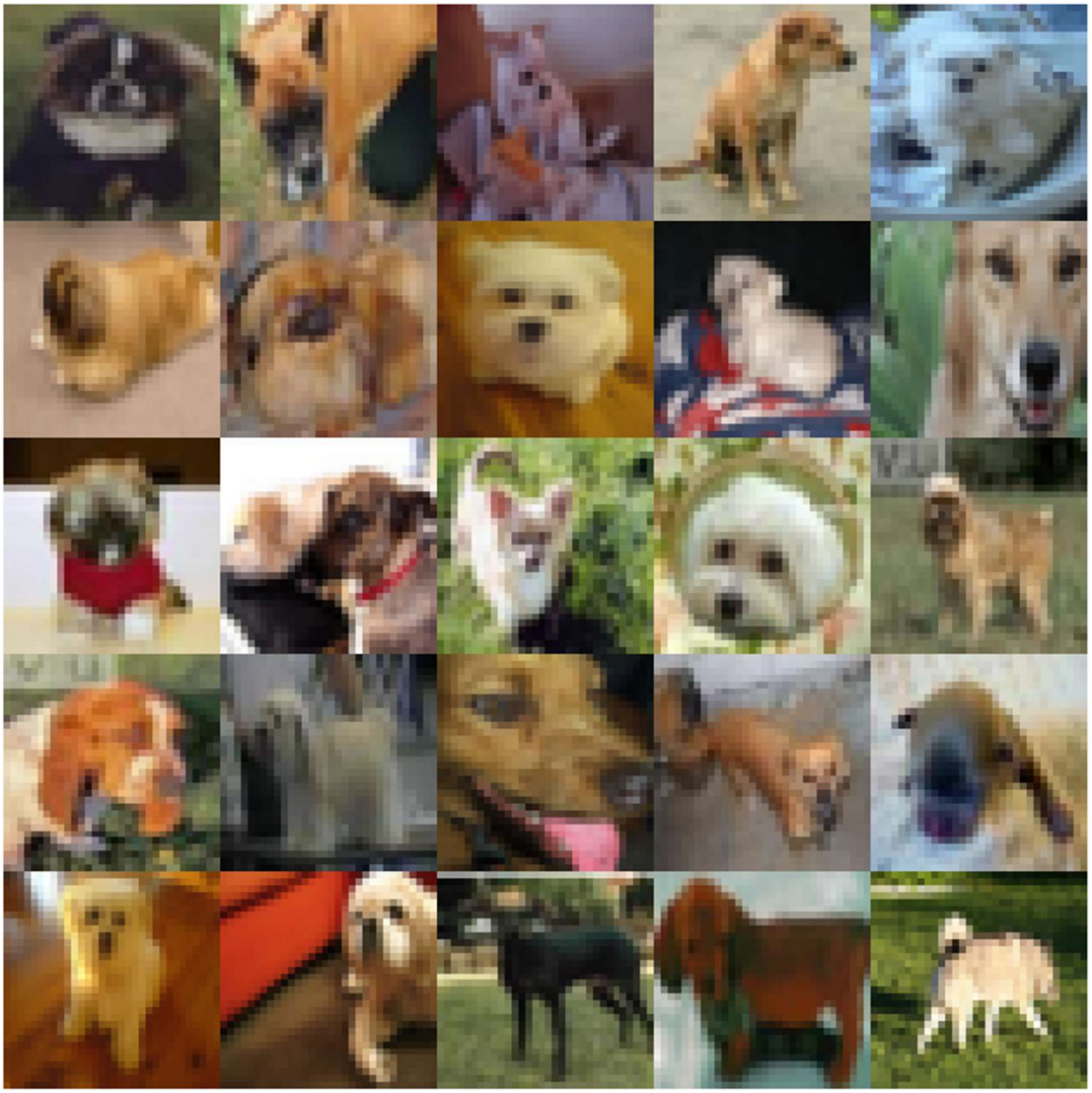


Image Classification


train
test
1.00
0.95
0.90
0.85
0.80
0.75
accuracy
0
20
40
60
80
100
value of ν


train
test
0.975
0.950
0.925
0.900
0.875
0.850
0.825
0.800
accuracy
10^3
of samples

least confident



most confident



```
QDA with nu=0: train=0.99, test=0.72
QDA with nu=1: train=0.98, test=0.78
QDA with nu=5: train=0.97, test=0.81
QDA with nu=10: train=0.96, test=0.82
QDA with nu=25: train=0.95, test=0.81
QDA with nu=50: train=0.94, test=0.81
QDA with nu=75: train=0.94, test=0.80
QDA with nu=100: train=0.93, test=0.79
Gaussian basis functions using 250 samples: train=0.81, test=0.81
Gaussian basis functions using 500 samples: train=0.84, test=0.82
Gaussian basis functions using 750 samples: train=0.86, test=0.84
Gaussian basis functions using 1000 samples: train=0.87, test=0.86
Gaussian basis functions using 2000 samples: train=0.91, test=0.86
Gaussian basis functions using 3000 samples: train=0.94, test=0.86
Gaussian basis functions using 5750 samples: train=0.99, test=0.88

Process finished with exit code 0
```